GeoHam: Symplectic-Regularised Feature Evolution for Efficient Remote Sensing Change Detection

Shailendra Dabral, Unmesh Khati, *Member, IEEE,* and Yash Raj *Student Member, IEEE*

***Abstract*—GeoHam is a geometric-regularisation framework inspired by Hamiltonian mechanics for remote-sensing change detection (CD). It replaces conventional feature subtraction with multi-scale Störmer-Verlet symplectic integration applied to *learned* feature dynamics. Unlike physics-informed neural networks that enforce PDE residuals from physical laws, GeoHam exploits the *mathematical structure* of Hamiltonian mechanics as an inductive bias on bitemporal feature trajectories: the symplectic 2-form is preserved at every leapfrog step, acting as a geometric regulariser that keeps distinct temporal states distinguishable. Three Feature Pyramid Network (FPN) levels (64×64, 32×32, 16×16) share a single integrator projected to a common 64-dimensional space; a raw absolute-difference skip at 128×128 preserves boundary detail. Integration steps per scale are selected by Bayesian hyperparameter optimisation. GeoHam achieves F1 of 91.20%, 95.89%, and 79.56% on LEVIR-CD, CDD, and CLCD respectively, with only 7.53 M parameters and 4.72 GFLOPs, 5.3× fewer GFLOPs than StarCD-Net without any transformer, state-space model, or cross-attention block.**

***Index Terms*—Change detection, Hamiltonian mechanics, symplectic integration, remote sensing, efficient deep learning, feature pyramid network.**

I. Introduction

REMOTE sensing change detection (CD) identifies land-cover modifications between co-registered bitemporal image pairs, supporting applications from urban expansion monitoring to disaster assessment [1]. Modern supervised methods have driven rapid progress through increasingly powerful comparison operators: FC-EF [2] learns early-fusion differences; BIT [3] uses semantic token transformers; ChangeFormer [4] applies hierarchical ViT at the cost of 41 M parameters and 202.8 GFLOPs; SNUNet-CD [5] adopts dense skip connections; and StarCD-Net [6] reaches competitive accuracy at 11 M parameters and 25 GFLOPs. Despite these advances, no existing method imposes geometric constraints on how features should *evolve* between time steps.

We observe that temporal feature evolution admits a natural dynamical-systems interpretation. If the T1 feature map represents the *configuration* $\mathbf{q}$ of a learned dynamical system, genuine structural changes correspond to high-energy transitions while nuisance variation (illumination, sensor noise, seasonal colour shift) induces low-energy perturbations. This distinction is reliable only when the integrator preserves geometric structure exactly. Standard Euler integration introduces $\mathcal{O}(\Delta t^2)$ drift per step; over multiple leapfrog iterations, this drift reaches the same order as the change signal, making genuine change and numerical error indistinguishable. Symplectic integrators, by contrast, preserve the phase-space 2-form $\omega = d\mathbf{q} \wedge d\mathbf{p}$ exactly at every step, bounding global energy error by a constant regardless of trajectory length [7], [8]. GeoHam uses the mathematical structure of the Störmer-Verlet scheme as an inductive bias on learned feature dynamics — a design we term *symplectic-regularised feature evolution*.

The main contributions of this work are as follows. 1) We present the first application of Störmer-Verlet symplectic integration as a geometric regulariser on learned bitemporal feature dynamics for change detection. 2) To deploy this at multiple spatial scales without a proportional increase in parameters, a single integrator with shared weights operates at three FPN levels in a common 64-dimensional projected space. 3) Integration steps per scale are selected by principled Bayesian optimisation using Optuna (15 trials × 20 epochs), consistently yielding a uniform $(4, 4, 4)$ step configuration. 4) We introduce a simplified boundary-aware focal loss. 5) The resulting model, GeoHam, achieves state-of-the-art efficiency with only 7.53 M parameters and 4.72 GFLOPs, surpassing StarCD-Net on LEVIR-CD by $+0.32$ pp F1 at 5.3× lower compute.

II. Related Work

A. Change Detection Architectures

Early convolutional methods such as FC-EF [2] concatenate bitemporal images before a UNet encoder. BIT [3] introduced semantic token transformers (3.55 M, 4.35 GFLOPs); ChangeFormer [4] applied hierarchical ViT [9] at the cost of 41 M parameters and 202.8 GFLOPs; SNUNet-CD [5] adopted dense skip connections (9.16 M, 78.49 GFLOPs); and StarCD-Net [6] achieved competitive accuracy at 11 M parameters and 25 GFLOPs. Unsupervised approaches have also been explored, including a Difference-in-Differences formulation for season-varying imagery [10] and random-forest-based methods using SPSA optimisation [11]. Specialised architectures for water-body change have been proposed using modified UNet designs [12]. GeoHam achieves 4.72 GFLOPs, substantially below all supervised baselines.

B. Physics-Inspired and Geometry-Aware Deep Learning

Physics-Informed Neural Networks (PINNs) [13] enforce differential equation residuals as soft constraints; GeoHam uses no PDE and no domain-specific physical law. Symplectic integrators [14], [15] are standard in molecular dynamics and celestial mechanics because they preserve the symplectic 2-form $\omega = d\mathbf{q} \wedge d\mathbf{p}$ exactly at every step [7]. GeoHam is the first

S. Dabral, U. Khati, and Y. Raj are with the Indian Institute of Technology Indore, Madhya Pradesh 453552, India (e-mail: ms2404121005@iiti.ac.in; unmesh.khati@iiti.ac.in; ms2404121001@iiti.ac.in).

work to apply symplectic geometric structure as an inductive bias on deep feature dynamics for CD.

III. Methodology

Notation. A bitemporal pair (I_1, I_2) consists of co-registered images at times $t_1 < t_2$. A shared EfficientNet-B2 encoder ϕ produces multi-scale features $\mathbf{f}_k^{(\ell)} = \phi_\ell(I_k)$ at four FPN levels $\ell \in \{0, 1, 2, 3\}$ with spatial resolutions 128×128, 64×64, 32×32, and 16×16. In the Störmer-Verlet scheme, $\mathbf{q}$ denotes the *configuration* (feature state), $\mathbf{p}$ the *conjugate momentum* (a learned latent encoding the direction and magnitude of expected feature change), and $H(\mathbf{q}, \mathbf{p})$ a *learned proxy functional* whose conservation is encouraged via a variance regularisation term.

A. Symplectic-Regularised Feature Evolution

GeoHam defines a learned Hamiltonian proxy on the feature space as a structural inductive bias. GeoHam does *not* constitute physics-informed machine learning in the sense of PINNs [13], which enforce PDE residuals derived from known physical laws. No PDE and no domain-specific physical law is used here. The Hamiltonian proxy $H(\mathbf{q}, \mathbf{p})$ is not a physical energy but a learned regularisation target. GeoHam exploits the *mathematical structure* of the Störmer-Verlet scheme which specifically, exact symplectic 2-form preservation-as an inductive bias on learned feature trajectories. We therefore term the approach *symplectic-regularised feature evolution*. There is no physical particle and no physical energy being conserved. The correspondence is:

- $\mathbf{q}_1 \leftarrow \mathbf{f}_1$: T1 features as initial configuration (scene state at t_1).
- $\mathbf{q}_2 \leftarrow \mathbf{f}_2$: T2 features as target configuration (scene state at t_2).
- $\mathbf{p}$: initialised by a 1×1 convolution from $\mathbf{q}_1$; a learned latent encoding the expected direction of scene change.
- $H(\mathbf{q}, \mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|^2 + \frac{1}{2}\|f_q(\mathbf{q})\|^2$: a proxy scalar used only for variance regularisation; f_q is a depthwise-separable convolution (3×3 depthwise + 1×1 pointwise, GroupNorm, GELU).

The gradient networks f_q and f_p approximate Hamilton's equations:

$$\dot{\mathbf{q}} \approx f_p(\mathbf{p}), \tag{1}$$

$$\dot{\mathbf{p}} \approx -f_q(\mathbf{q}). \tag{2}$$

f_p is a 1×1 pointwise convolution (GroupNorm, GELU) so that momentum propagates channel-wise. Both networks operate on the shared 64-dimensional projected feature space.

B. Störmer-Verlet Leapfrog Integration

Given step count N and $\Delta t = 1/N$, the Störmer-Verlet scheme executes:

$$\mathbf{p}_{1/2} = \mathbf{p}_0 - \tfrac{\Delta t}{2} f_q(\mathbf{q}_0), \tag{3}$$

$$\mathbf{q}_n = \mathbf{q}_{n-1} + \Delta t\, f_p(\mathbf{p}_{n-1/2}) \quad \text{for } n = 1 \ldots N, \tag{4}$$

$$\mathbf{p}_{n+1/2} = \mathbf{p}_{n-1/2} - \Delta t\, f_q(\mathbf{q}_n) \quad \text{for } n = 1 \ldots N-1, \tag{5}$$

$$\mathbf{p}_N = \mathbf{p}_{N-1/2} - \tfrac{\Delta t}{2} f_q(\mathbf{q}_N). \tag{6}$$

The staggered half-kicks in (3) and (6) make this scheme *symplectic*: the discrete flow exactly preserves $\omega = d\mathbf{q} \wedge d\mathbf{p}$ at every step, guaranteeing that distinct initial configurations remain distinguishable after integration. This is the central geometric property exploited by GeoHam: two image pairs with genuinely different change patterns always produce distinguishably different divergence maps, a guarantee absent from Euler methods. After N steps, the change divergence at this scale is:

$$\mathbf{D} = |\mathbf{q}_N - \mathbf{q}_2|. \tag{7}$$

C. Multi-Scale Integration with Shared Weights

Processing bitemporal differences at multiple spatial resolutions is critical: fine scales resolve building edges and crop parcel boundaries, while coarse scales capture large-area land-cover transitions. To cover three FPN levels without tripling parameters, all scales are projected to a shared 64-dimensional space before integration.

For each FPN level $\ell \in \{1, 2, 3\}$ with C_ℓ channels, a learned projector π_ℓ maps features to the shared space:

$$\hat{\mathbf{f}}_k^{(\ell)} = \text{GELU}(\text{BN}(\pi_\ell(\mathbf{f}_k^{(\ell)}))) \in \mathbb{R}^{64 \times H_\ell \times W_\ell}. \tag{8}$$

Because the co-registered image pair is processed by the same shared encoder, same-level temporal features already lie on an identical $H_\ell \times W_\ell$ grid; the 1×1 projectors change only the channel dimension and perform no spatial resampling. A single Störmer-Verlet integrator with *identical weights across all three scales* then applied to $(\hat{\mathbf{f}}_1^{(\ell)}, \hat{\mathbf{f}}_2^{(\ell)})$ with N_ℓ steps (determined by Bayesian optimisation; see Section IV-B), yielding divergences $\mathbf{D}^{(1)}, \mathbf{D}^{(2)}, \mathbf{D}^{(3)}$. Weight sharing reduces parameters versus three separate integrators and acts as a scale-invariant structural prior. A per-scale depthwise refinement convolution recovers scale-specific detail after integration.

Scale-0 skip. A raw absolute-difference skip at 128×128 is projected to 32 channels without integration, preserving maximum boundary resolution: $\mathbf{S}^{(0)} = \pi_0(|\mathbf{f}_1^{(0)} - \mathbf{f}_2^{(0)}|)$.

D. FPN Decoder

The FPN decoder fuses scales coarse-to-fine. Before each fusion, the coarser map is bilinearly resized to the exact height and width of the next lateral map, concatenated with it along the channel dimension, and processed by two 3×3 Conv–BN–SiLU layers. The spatial sequence is $16^2 \to 32^2 \to 64^2 \to 128^2$; no lateral feature is downsampled. A final bilinear resize to 256^2 and a Conv-BN-SiLU-Conv head produces the prediction. Full architecture is illustrated in (Fig. 1); the complete formulation is given in Supplementary S5.

E. Training Objective

Boundary map. From ground-truth mask $\mathbf{y}$, a morphological gradient boundary map is computed as:

$$\mathbf{B}(\mathbf{y}) = \text{dilate}(\mathbf{y}) - \text{erode}(\mathbf{y}), \tag{9}$$

implemented via max-pooling and min-pooling with radius 3.

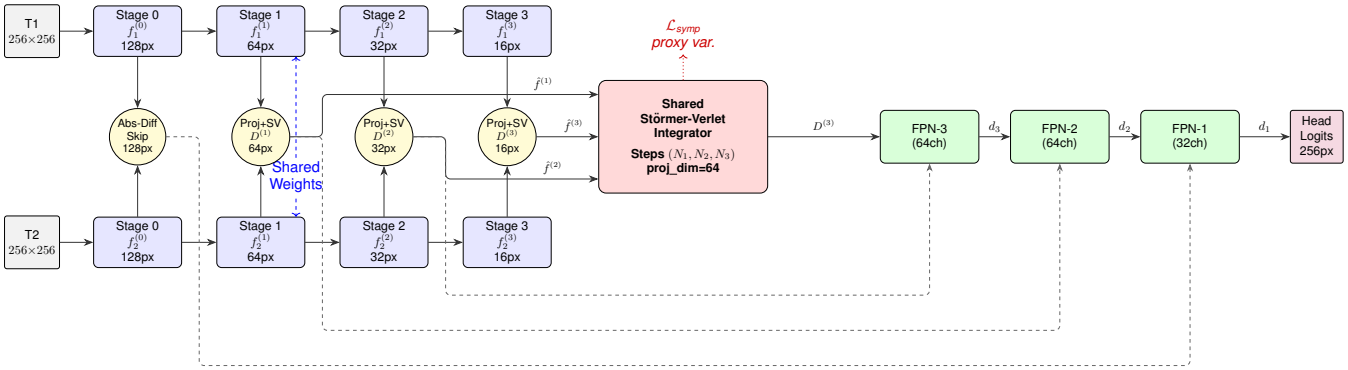


Fig. 1. GeoHam architecture. A shared EfficientNet-B2 encoder produces features at four FPN levels. Levels 1–3 are projected to 64-channel space and processed by the *shared* Störmer-Verlet integrator with (N_1, N_2, N_3) steps. Level 0 (128×128) provides a raw absolute-difference skip without integration, preserving maximum boundary resolution. The FPN decoder fuses coarse-to-fine; the dotted red path carries proxy Hamiltonian values used only as a training signal for symplectic regularisation $\mathcal{L}_{\text{symp}}$.

TABLE I
PER-DATASET HYPERPARAMETERS SELECTED BY OPTUNA (15 TRIALS × 20 EPOCHS EACH).

Dataset	λ_{bnd}	λ_{symp}	α_{focal}	LR	Steps
LEVIR-CD	0.528	0.002	0.641	1.03e-4	(4,4,4)
CDD	0.481	0.010	0.681	2e-4	(4,4,4)
CLCD	0.202	0.042	0.660	2.7e-4	(4,4,4)

Boundary focal loss. Boundary pixels receive $1+\beta$ times the baseline focal weight:

$$\mathcal{L}_{\text{bnd}} = \mathbb{E}_x\left[\left(1+\beta\,\mathbf{B}(\mathbf{y})(x)\right)\cdot\mathcal{L}_{\text{focal}}(\hat{y}(x), y(x))\right], \qquad (10)$$

with $\beta = 3$ (boundary pixels receive $4\times$ weight).

Symplectic regularisation. Recording the proxy Hamiltonian H_n at each leapfrog step, we penalise its variance across the trajectory:

$$\mathcal{L}_{\text{symp}} = \text{Var}\left(\{H_n\}_{n=1}^{N_{\text{total}}}\right), \qquad (11)$$

where $N_{\text{total}} = \sum_\ell N_\ell = 12$ for the $(4,4,4)$ configuration. This prevents f_q and f_p from collapsing to zero and enforces mutual consistency between the two gradient networks.

Full objective.

$$\mathcal{L} = \mathcal{L}_{\text{focal-dice}} + \lambda_{\text{bnd}}\,\mathcal{L}_{\text{bnd}} + \lambda_{\text{symp}}\,\mathcal{L}_{\text{symp}}, \qquad (12)$$

where $\mathcal{L}_{\text{focal-dice}}$ combines focal loss [16] ($\gamma = 2$) with Dice loss [17]. Loss weights and per-dataset values are given in Table I.

IV. EXPERIMENTS

A. Datasets

The evaluation utilizes three distinct change detection datasets: LEVIR-CD [18], comprising 9,168 total bitemporal image pairs (256×256 px) focused on building construction and demolition; CDD [19], which captures seasonal illumination variance across vegetation, urban, and agricultural scenes; and CLCD [20], a highly challenging cropland dataset containing 1,000 pairs (512×512 px, centre-cropped to 256×256 px for training) that features fine-grained intra-class variations in agricultural land.

Official train/val/test splits are used for all datasets.

B. Implementation Details

Training uses AdamW [21] with OneCycleLR [22] (100 epochs, 15% warmup, batch 16 for all datasets). Mixed-precision training [23] and dynamic Exponential Moving Average (EMA, $\tau \in [0.99, 0.999]$) are applied. Test-Time Augmentation (TTA) averages predictions over horizontal flip, vertical flip, and the original orientation.

Dataset-specific hyperparameters (λ_{bnd}, λ_{symp}, α_{focal}, learning rate, and integration steps N_1, N_2, N_3) are selected by Bayesian optimisation using Optuna [24] with 15 trials of 20 epochs each on the validation set. The binary classification threshold is swept over $\{0.30, 0.35, \ldots, 0.80\}$ on the *validation* set and frozen before test evaluation. Parameters and GFLOPs are measured using THOP [25] at 256×256 input. Inference latency is measured on an NVIDIA Tesla T4 (16 GB VRAM) with 50 warmup and 200 timed runs using `torch.cuda.synchronize()` for accurate GPU timing.

Metrics. We report Precision (P), Recall (R), F1, Intersection over Union (IoU), and Overall Accuracy (OA) on the change class. Let TP, FP, TN, FN denote true/false positives/negatives at the pixel level. Then

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad R = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad \text{F1} = \frac{2PR}{P+R},$$
$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}}, \quad \text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}}.$$

Table III compares GeoHam against leading baselines on LEVIR-CD. GeoHam achieves F1 = 91.20% and IoU = 83.82% with only 4.72 GFLOPs and 7.53 M parameters, **surpassing StarCD-Net by +0.32 pp F1** at 5.3× fewer GFLOPs, and improving over BIT by +1.89 pp at comparable cost (4.72 vs. 4.35 GFLOPs). Relative to BIT, $\Delta\text{F1}/\text{GFLOPs} \times 1000$ is 400.4 for GeoHam and 62.8 for StarCD-Net. Table IV reports F1 = 95.89% on CDD, exceeding StarCD-Net by +0.13 pp, and F1 = 79.56% (IoU = 66.06%) on CLCD. Performance across building, seasonal, and cropland changes supports the cross-domain applicability of the learned geometric prior.

C. Ablation Study

Table VI isolates each design choice on the LEVIR-CD official test set. All variants use 100-epoch training, seed 42, and a frozen validation threshold.

TABLE II
COMPARISON ON THE CDD TEST SET WITH GFLOPS AND PARAMS. BEST PER COLUMN IN **BOLD**. FULL BENCHMARK WITH LATENCY, THROUGHPUT, AND MEMORY IS PROVIDED IN SUPPLEMENTARY S2

Method	F1 (%)	IoU (%)	Pr (%)	Re (%)	Params (M)	GFLOPs
FC-EF [2]	86.47	76.17	89.91	83.28	1.87	8.05
DTCDSCN [26]	84.40	84.26	89.42	80.64	41.07	40.78
SNUNet-CD [5]	91.60	84.51	93.57	89.72	27.44	27.44
BIT [3]	91.68	84.65	93.32	92.29	3.55	4.35
ChangeFormer [4]	88.65	88.01	90.84	86.77	41.00	202.80‡
StarCD-Net [6]	95.76	91.87	96.28	95.25	11.00	25.00†
GeoHam (Ours)	**95.89**	**92.10**	**96.39**	**95.39**	**7.53**	**4.72**

†GFLOPs from original publication. ‡Published GFLOPs; re-measured at 405.58 on Tesla T4 in Supplementary S2 due to implementation differences.

TABLE III
COMPARISON ON THE LEVIR-CD TEST SET. BEST PER COLUMN IN **BOLD**.

Method	F1 (%)	IoU (%)	Pr (%)	Re (%)	OA (%)
FC-EF [2]	83.40	71.53	86.91	80.17	98.39
DTCDSCN [26]	87.67	78.05	88.53	86.83	98.77
BIT [3]	89.31	80.68	89.29	89.37	98.92
SNUNet-CD [5]	88.16	78.83	89.18	87.17	98.82
ChangeFormer [4]	90.40	82.65	91.83	89.01	99.00
StarCD-Net [6]	90.88	83.28	91.37	90.40	99.08
GeoHam (Ours)	**91.20**	**83.82**	**91.32**	**91.09**	**99.10**

TABLE IV
FULL METRICS EVALUATION OF GEOHAM ACROSS ALL DATASETS.

Dataset	F1 (%)	IoU (%)	Pr (%)	Re (%)	OA (%)
LEVIR-CD	91.20	83.82	91.32	91.09	99.10
CDD	95.89	92.10	96.39	95.39	98.98
CLCD	79.56	66.06	80.69	78.55	97.00

TABLE V
ROBUSTNESS ANALYSIS OF GEOHAM ON THE CLCD DATASET ACROSS DIFFERENT RANDOM SEEDS ALONG WITH COMPARISON TO EXISTING METHODS.

Method	F1	Pr	Re	OA
GeoHam(S1)	79.81	81.19	78.81	97.05
GeoHam(S2)	79.47	81.36	77.68	97.01
GeoHam(S3)	79.40	78.98	79.83	96.92
GeoHam(S4)	79.56	80.90	78.26	97.01
GeoHam(S5)	79.58	81.04	78.17	97.02
GeoHam($\mu \pm \sigma$)	$\mathbf{79.5 \pm 0.1}$	80.6 ± 0.9	78.5 ± 0.7	97.00 ± 0.05
SNUNet	69.50	69.14	69.87	92.89
BIT	71.67	75.65	68.09	93.76
CroplandCDNet	76.04	76.46	75.63	94.47

Integration step configuration. The best Optuna trial for each dataset selected the uniform $(4, 4, 4)$ configuration. Although its step count is fixed, adaptation arises from the scale-specific encoder features, projectors, and refiners: the 64×64 level retains local boundaries, whereas the 32×32 and 16×16 levels supply broader context. Four steps give every scale the same integration horizon ($\Delta t = 0.25$) and numerical depth, while shared weights promote scale-consistent evolution. Thus $(4, 4, 4)$ is a robust empirical operating point across scenarios, rather than an input-adaptive rule.

TABLE VI
ABLATION STUDY ON LEVIR-CD (FIXED TEST AND VALIDATION SET).

Variant	F1 (%)	IoU (%)
Integration step configuration		
Steps $(1, 1, 1)$	90.31	82.34
Steps $(2, 3, 4)$	90.88	83.20
Steps $(3, 3, 3)$	90.41	82.52
Integration scheme		
Euler integration	90.14	82.04
GeoHam (full, 4,4,4)	**91.20**	**83.82**

Symplectic vs. Euler integration. Replacing Störmer-Verlet with simultaneous Euler reduces F1 by 1.06 pp. With architecture, hyperparameters, and seed fixed, accumulated Euler drift causes the divergence map (7) to mix structural change with numerical artefacts. **Loss components.** Additional details regarding the isolated ablation of the focal-dice, boundary, and symplectic regularisation loss terms are provided in Supplementary S1 (Table S1).

D. Qualitative Results

Fig. 2 shows representative predictions on all three datasets. GeoHam accurately delineates building boundaries on LEVIR-CD (Pr = 91.32%, Re = 91.09%), captures large-scale seasonal transitions on CDD (Pr = 96.39%, Re = 95.39%), and detects cropland boundaries on CLCD (Pr = 80.69%, Re = 78.5%)) despite high intra-class variation. Remaining errors are concentrated at fine-scale building edges and crop parcel boundaries. Error maps: green = True Positive, red = False Positive, blue = False Negative.

V. CONCLUSION

GeoHam is the first change detection model to apply multi-scale Störmer-Verlet symplectic integration as a geometric regulariser on learned bitemporal feature dynamics. By treating feature evolution as a Hamiltonian-inspired dynamical system and integrating at three FPN scales with shared weights, GeoHam produces divergence maps that separate structural change from appearance noise without transformer blocks, state-space models, or domain-specific pre-processing. The integration step count $(4, 4, 4)$ is determined by Bayesian optimisation and experiment on Levir-CD rather than just manual tuning. Evaluated on LEVIR-CD, CDD, and CLCD,

GeoHam achieves 91.20%, 95.89%, and 79.56% F1 respectively, surpassing all compared methods on LEVIR-CD and CDD with only 7.53 M parameters and

4.72 GFLOPs. On CLCD, the higher precision than recall (80.69% vs. 78.55%) suggests the model errors toward missing fine crop-boundary changes; domain-adapted agricultural pre-training is a natural avenue for future work.

ACKNOWLEDGMENT

The authors thank Kaggle for access to GPU resources used for benchmarking, and the anonymous reviewers for their constructive feedback.

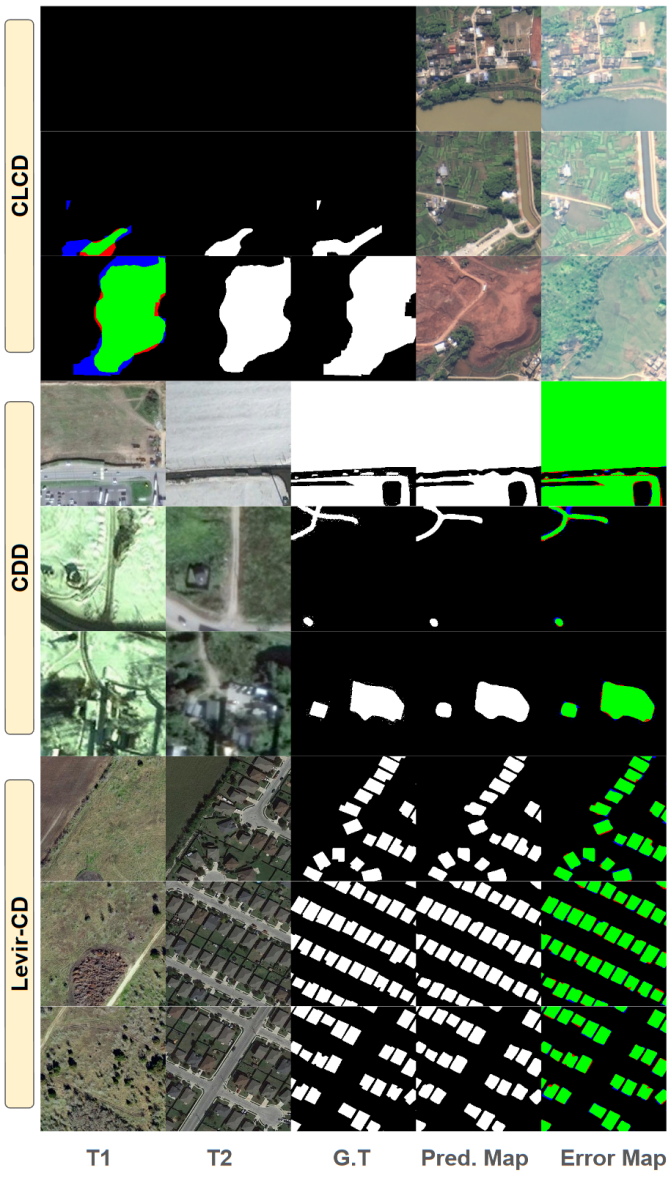


Fig. 2. Qualitative results on LEVIR-CD (rows 1-2), CDD (rows 3-4), and CLCD (rows 5–6). Columns: T1, T2, Ground Truth, Prediction, Error map (green = TP, red = FP, blue = FN).

REFERENCES

[1] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE transactions on geoscience and remote sensing*, vol. 60, pp. 1–16, 2021.

[2] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional siamese networks for change detection," in *2018 25th IEEE international conference on image processing (ICIP)*. IEEE, 2018, pp. 4063–4067.

[3] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2021.

[4] W. G. C. Bandara and V. M. Patel, "A transformer-based siamese network for change detection," in *IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium*. IEEE, 2022, pp. 207–210.

[5] S. Fang, K. Li, J. Shao, and Z. Li, "Snunet-cd: A densely connected siamese network for change detection of vhr images," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2021.

[6] Q. Hu, Y. Wang, Y. Zhang, W. Yang, F. Xu, and G.-S. Xia, "Star-cd: Style-aligned remote sensing change detection with appearance-relation modeling," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–15, 2025.

[7] B. Leimkuhler and S. Reich, *Simulating hamiltonian dynamics*. Cambridge university press, 2004, no. 14.

[8] E. Hairer, C. Lubich, and G. Wanner, "Geometric numerical integration illustrated by the störmer–verlet method," *Acta numerica*, vol. 12, pp. 399–450, 2003.

[9] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[10] Y. Yuan, X. Chen, K. Tang, and J. Chen, "A "difference-in-differences"-based method for unsupervised change detection in season-varying images," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2024.

[11] A. Musku, H. Padala, S. Paul, and D. K. Mahapatra, "Change detection in remote sensing optical images using a spsa-based improved random forest algorithm," *IEEE Access*, vol. 13, pp. 201 264–201 276, 2025.

[12] H. Padala, A. Musku, S. Paul, and D. K. Mahapatra, "A novel waterbody change detection in optical remote sensing data using a modified u-net," *International Journal of Remote Sensing*, vol. 47, no. 2, pp. 928–958, 2026.

[13] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational physics*, vol. 378, pp. 686–707, 2019.

[14] C. Störmer, "Sur les trajectoires des corpuscules électrisés dans l'espace sous l'action du magnétisme terrestre, avec application aux aurores boréales," *Le Radium*, vol. 9, no. 11, pp. 395–399, 1912.

[15] L. Verlet, "Computer" experiments" on classical fluids. i. thermodynamical properties of lennard-jones molecules," *Physical review*, vol. 159, no. 1, p. 98, 1967.

[16] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980–2988.

[17] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *2016 fourth international conference on 3D vision (3DV)*. IEEE, 2016, pp. 565–571.

[18] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote sensing*, vol. 12, no. 10, p. 1662, 2020.

[19] M. Lebedev, Y. V. Vizilter, O. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 42, pp. 565–571, 2018.

[20] M. Liu, Z. Chai, H. Deng, and R. Liu, "A cnn-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 4297–4306, 2022.

[21] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017.

[22] L. N. Smith and N. Topin, "Super-convergence: Very fast training of neural networks using large learning rates," in *Artificial intelligence and machine learning for multi-domain operations applications*, vol. 11006. SPIE, 2019, pp. 369–386.

[23] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh *et al.*, "Mixed precision training," *arXiv preprint arXiv:1710.03740*, 2017.

[24] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2019, pp. 2623–2631.

[25] L. Zhu, "THOP: PyTorch-OpCounter," https://github.com/Lyken17/pytorch-OpCounter, 2020, accessed: March 20, 2026.

[26] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep siamese convolutional network model," *IEEE Geoscience and Remote Sensing Letters*, vol. 18, no. 5, pp. 811–815, 2020.